

Computational modeling of neural populations

Part III: Neural fields — Amari model

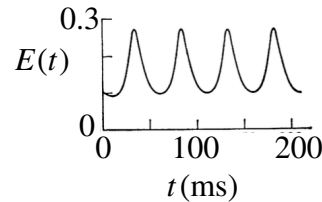
Why spatial extension?

- Localized Wilson-Cowan focuses on only temporal dynamics.

- Resting state
- Limit cycles

⋮

Limit cycles:



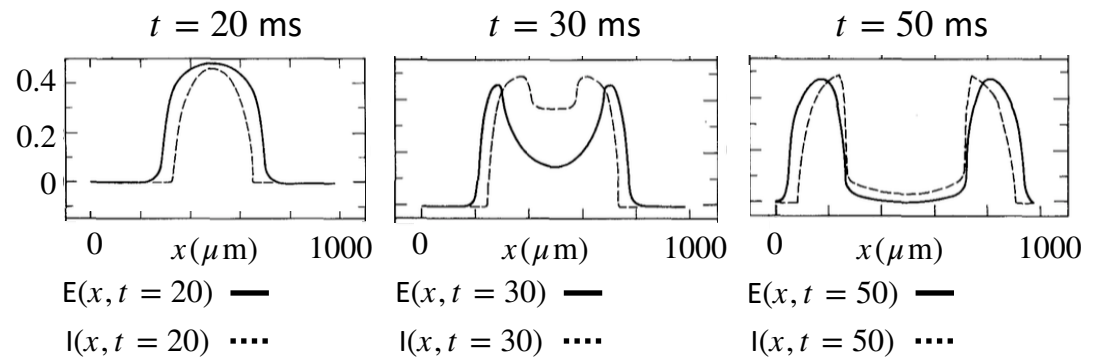
Wilson-Cowan 1972
Biophysical J

- Spatially extended Wilson-Cowan focuses on spatiotemporal dynamics.

- **Spatially** inhomogeneous steady states
- **Traveling** waves

⋮

Traveling waves:



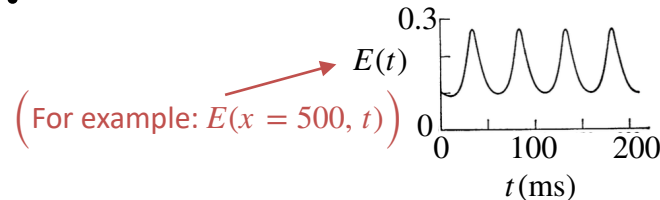
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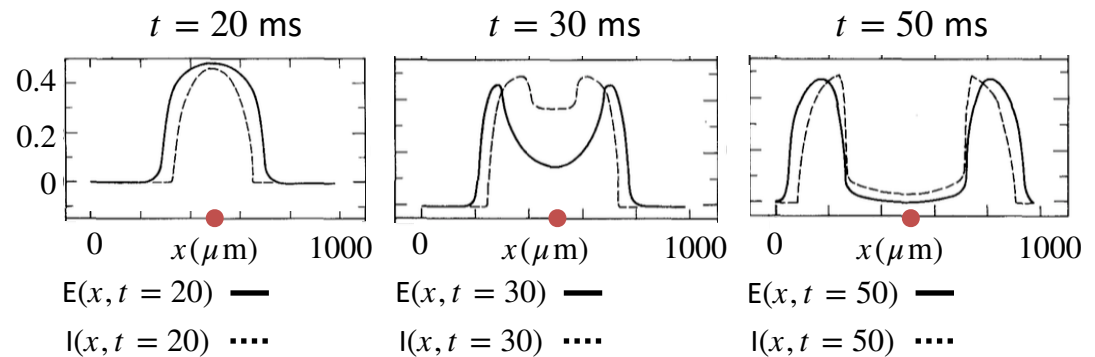
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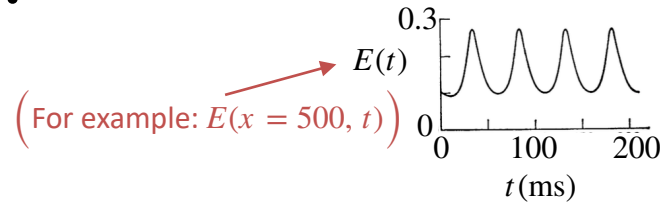
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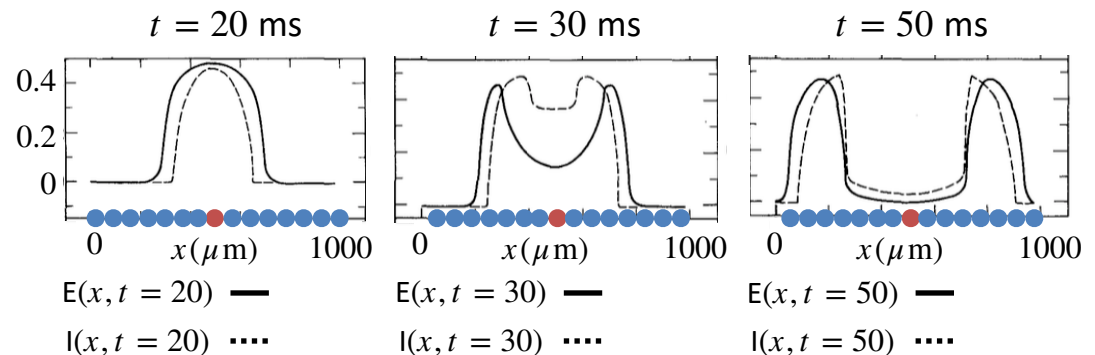
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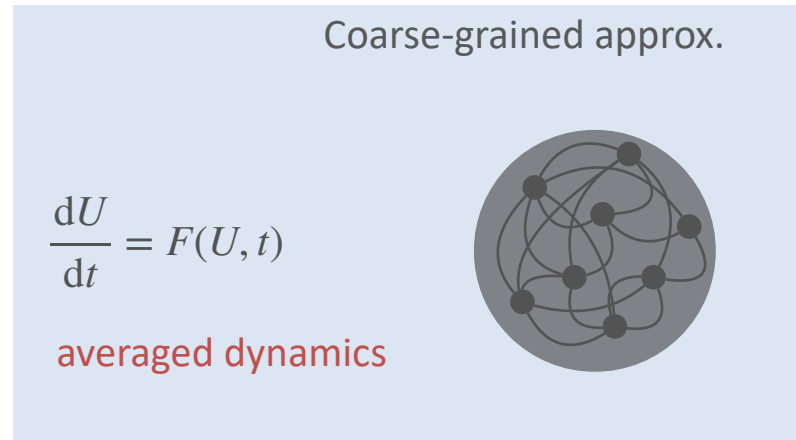
- **Spatially** inhomogeneous steady states
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⋮

Traveling waves:



Spatial extension allows us to consider the neurons which are not necessarily in close proximity.



Localized neural field: $\frac{dU}{dt} = F(U, t), \quad U(t)$

- Neurons in close proximity
- No spatial dependence
- No concept of connectivity

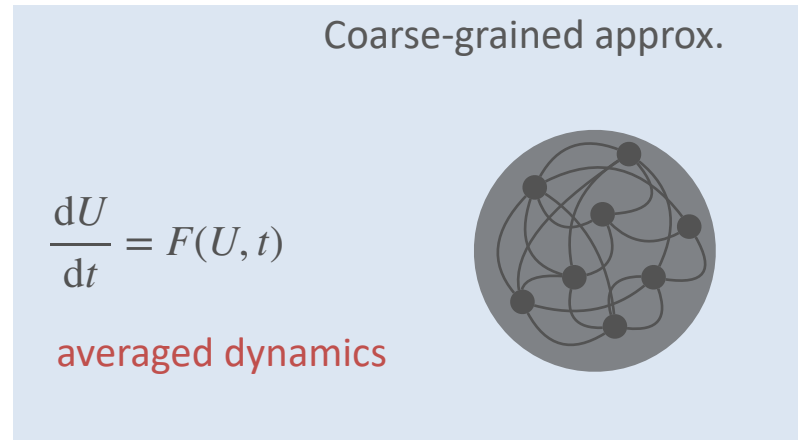
- Wilson-Cowan '72

Spatially extended neural field: $\frac{\partial U}{\partial t} = F(U, x, t), \quad U(x, t)$

- Spatially distributed neurons
- Spatial dependence
- Distance-based connectivity

- Wilson-Cowan '73

- Amari '77



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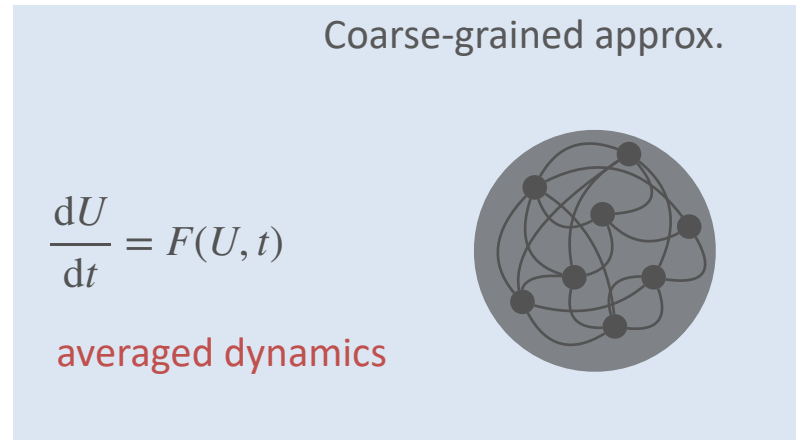
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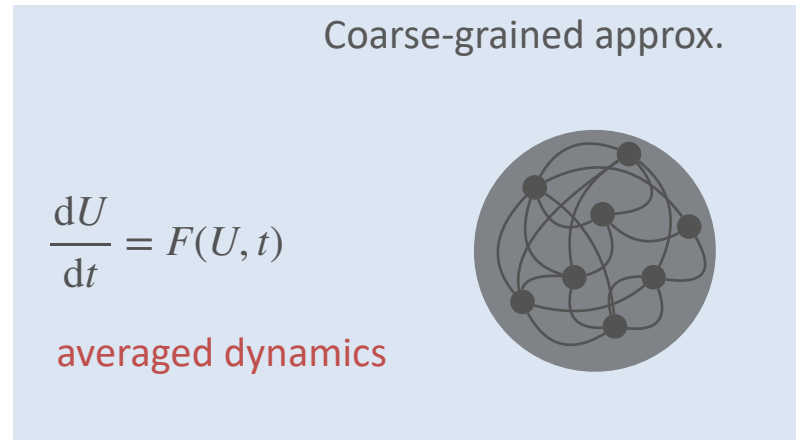
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Spatially extended neural field: $\frac{\partial U}{\partial t} = F(U, x, t), \quad U(x, t)$

- Spatially distributed neurons
- Spatial dependence
- Distance-based connectivity **encoding both excitation and inhibition**

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- ▶ **Single equation representing excitatory activity experiencing both exc. and inh. effects**

Modeling assumptions

1. Homogeneity

- Connectivity kernel is time independent, it does not change in time.
- Connectivity kernel depends only on the distance between the interacting, not on the position of neurons.

2. Short-range (local) excitation, long range (lateral) inhibition

- Neurons excite strongly their close neighbors (local positive feedback).
- They inhibit neurons further away weakly but broadly (global competition).
- Such connectivity is found, e. g. in the visual cortex: neurons tuned to similar orientations excite each other, whereas the neurons tuned to different orientations inhibit each other via lateral inhibition which can lie across larger cortical distances.
 - This enhances contrast, therefore provide a better selectivity to the perceptual features of the input

3. Population neural activity can be represented in terms of the average membrane potential of the population.

Spatially extended Wilson-Cowan model

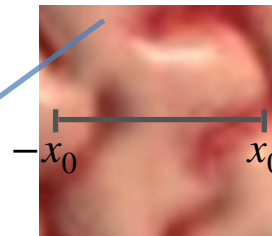
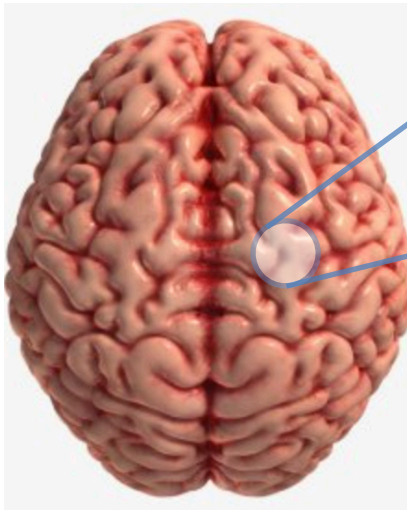
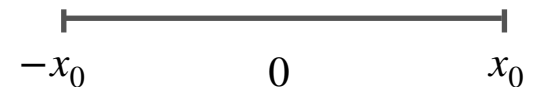
Wilson-Cowan, Kybernetik, 1973

$$\begin{aligned}\mu \frac{\partial u(x, t)}{\partial t} &= -u(x, t) + (s(u) \circledast \omega)(x, t) + h + P(x, t) \\ &= -u(x, t) + \int_{-x_0}^{x_0} s(u(y)) \omega(x - y) dy + h + P(x, t)\end{aligned}$$

$x \in [-x_0, x_0], \quad t \in [0, T]$

with boundary cond'ns
initial cond'ns

$[-x_0, x_0] \subset \mathbb{R}$: spatial interval cortical layer



- Cortical surface is assumed to be isotropic:
 - Direction has no impact on the connectivity,
 - Only distance matters in the connectivity.

$$\mu \frac{\partial u(x, t)}{\partial t} = -u(x, t) + (s(u) \circledast \omega)(x, t) + h + P(x, t) \quad x \in [-x_0, x_0], \quad t \in [0, T]$$

$$= -u(x, t) + \int_{-x_0}^{x_0} s(u(y)) \omega(x - y) dy + h + P(x, t) \quad \text{with boundary cond'ns}$$

initial cond'ns

$[-x_0, x_0] \subset \mathbb{R}$: spatial interval cortical layer

T : positive constant final time

$u : [-x_0, x_0] \times [0, T] \rightarrow \mathbb{R}$ average membrane potential of excitatory neurons
located at x and at instant t

s : nonlinear function population transfer function

h : constant threshold stimulus

$P : [-x_0, x_0] \times [0, T] \rightarrow \mathbb{R}$ time-dependent external stimulus

ω : nonlinear function connectivity kernel

μ : positive constant time constant

Amari model

$$\begin{aligned}\mu \frac{\partial u(x, t)}{\partial t} &= -u(x, t) + (s(u) \circledast \omega)(x, t) + h + P(x, t) \\ &= -u(x, t) + \int_{-x_0}^{x_0} s(u(y)) \omega(x - y) dy + h + P(x, t)\end{aligned}$$

- No explicit representation of excitatory and inhibitory activity.
- Connectivity kernel ω encodes both excitatory and inhibitory connections.
- Order of convolution and transfer function: first transfer function output, then convolution.
- Neural activity is represented in terms of average membrane potential.

Spatially-extended Wilson-Cowan model

$$\begin{aligned}\mu \frac{\partial \bar{E}(x, t)}{\partial t} &= -\bar{E}(x, t) + (1 - \bar{E}(x, t)) S_e \left((\bar{E} \circledast \omega_{ee})(x, t) - (\bar{I} \circledast \omega_{ei})(x, t) + P(x, t) \right) \\ \mu \frac{\partial \bar{I}(x, t)}{\partial t} &= -\bar{I}(x, t) + (1 - \bar{I}(x, t)) S_i \left((\bar{E} \circledast \omega_{ie})(x, t) - (\bar{I} \circledast \omega_{ii})(x, t) + Q(x, t) \right)\end{aligned}$$

- Both excitatory and inhibitory activities are represented explicitly.
- Inhibitory connections are encoded through the "-" sign, ω provides only the strength.
- Order of convolution and transfer function: first convolution, then transfer function output.
- Neural activity is represented in terms of proportion of active (spiking) neurons.

Amari model

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- No explicit representation of excitatory and inhibitory activity.
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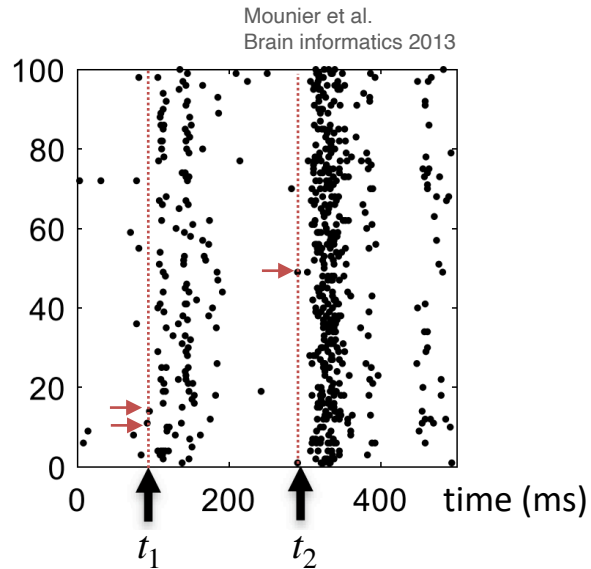
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Neural activity in Wilson-Cowan

neuron index

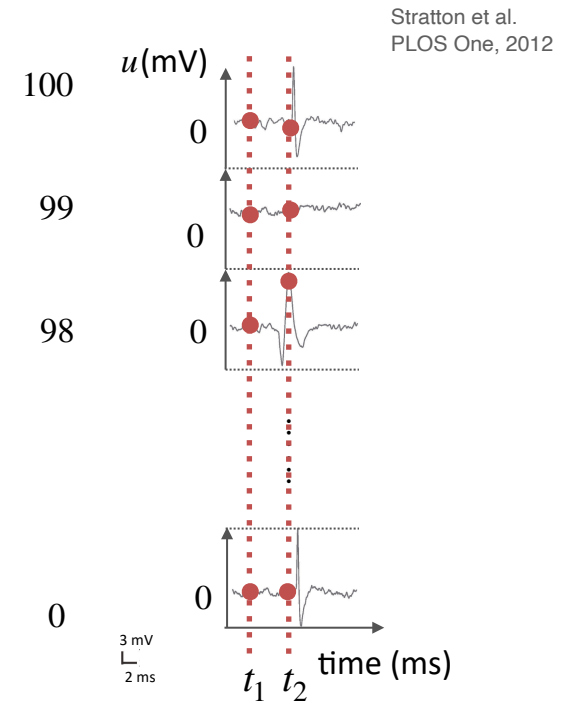


$$\bar{E}(x_0, t_1) = 0.02$$

$$\bar{E}(x_0, t_2) = 0.01$$

Neural activity in Amari

neuron index



$$u(x_0, t_1) = 12 \text{ mV}$$

$$u(x_0, t_2) = 28 \text{ mV}$$

$$\begin{aligned} \mu \frac{\partial u(x, t)}{\partial t} &= -u(x, t) + (s(u) \circledast \omega)(x, t) + h + P(x, t) & x \in [-x_0, x_0], \quad t \in [0, T] \\ &= -u(x, t) + \int_{-x_0}^{x_0} s(u(y)) \omega(x - y) dy + h + P(x, t) & \text{with boundary cond'ns} \\ & & \text{initial cond'ns} \end{aligned}$$

$[-x_0, x_0] \subset \mathbb{R}$: spatial interval cortical layer

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$u : [-x_0, x_0] \times [0, T] \rightarrow \mathbb{R}$ average membrane potential of excitatory neurons
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$$\mu \frac{\partial u(x, t)}{\partial t} = -u(x, t) + \int_{-\infty}^{\infty} s(u(y)) \omega(x - y) dy + h + P(x, t)$$

- Negative linear feedback: It ensures that the average membrane potential decays to the threshold value h in the absence of any synaptic input and external stimulus.
- Synaptic interaction: It models the interactions between the neurons.
- Threshold stimulus: It determines the resting state value. It is chosen generally as a negative value.
 - Generally negative since neural activity is represented in terms of membrane potentials.
- Time-dependent external stimulus

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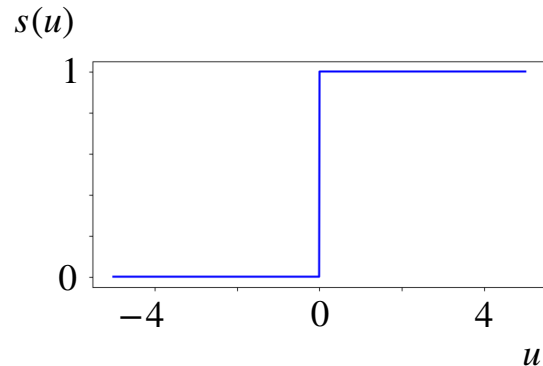
Amari model: synaptic interactions

Amari, Bio. Cybernetics, 1977

$$\mu \frac{\partial u(x, t)}{\partial t} = -u(x, t) + \int_{-\infty}^{\infty} s(u(y)) \omega(x - y) dy + h + P(x, t)$$

Transfer function

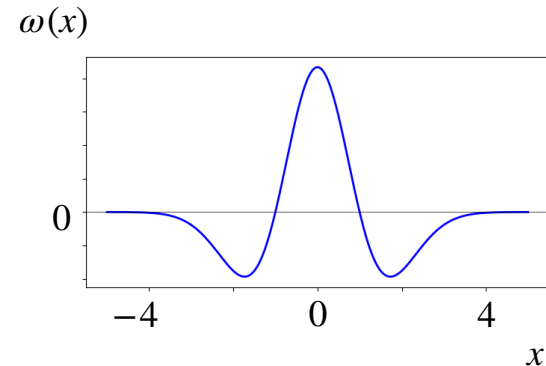
$$s(u) := \begin{cases} 0, & u \leq 0 \\ 1, & u > 0 \end{cases}$$



Heaviside step

Connectivity kernel

$$\omega(x) := \frac{2}{\sqrt{3} \sigma \pi^{1/4}} \left(1 - \frac{x^2}{\sigma^2}\right) e^{-\frac{x^2}{2\sigma^2}}$$



Mexican hat

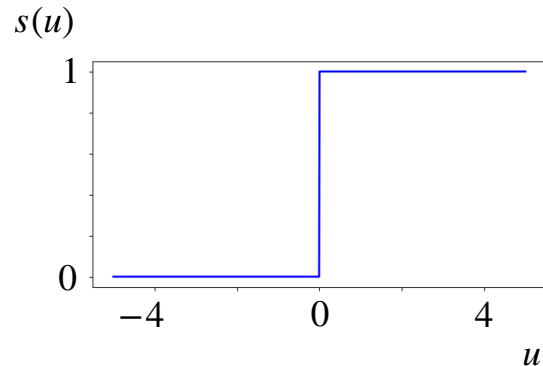
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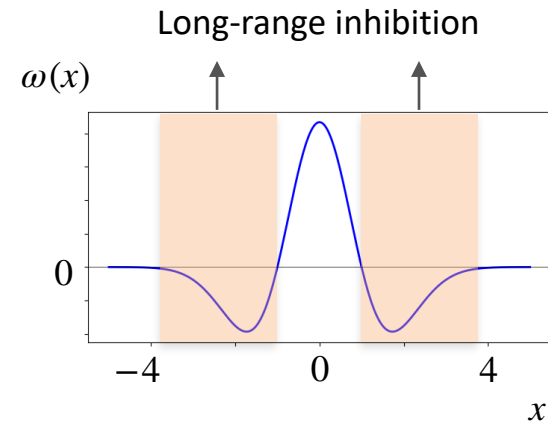
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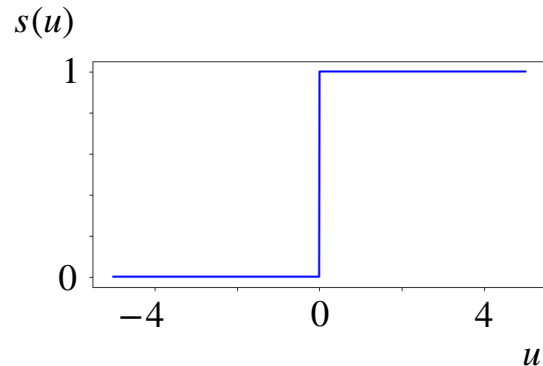
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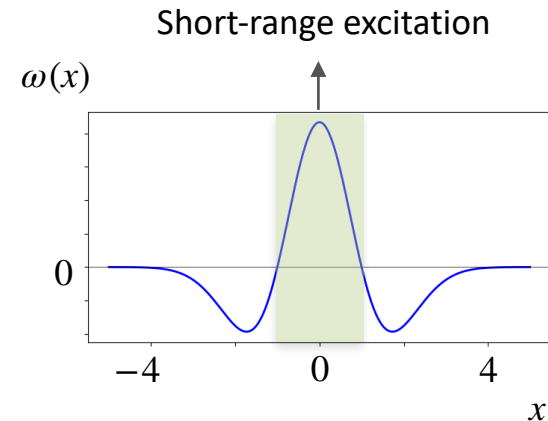
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Short-range excitation

Mexican hat

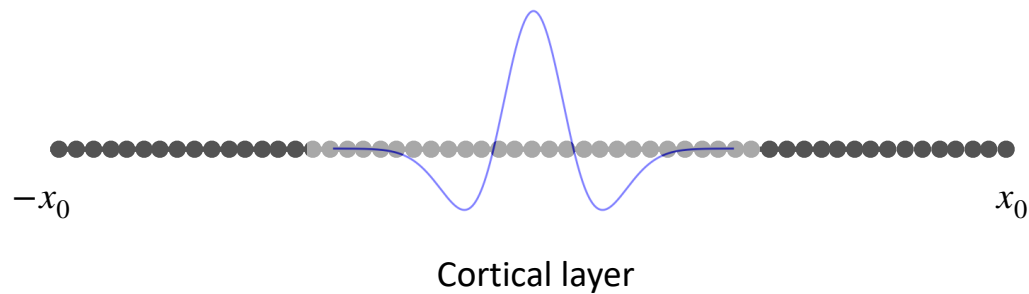
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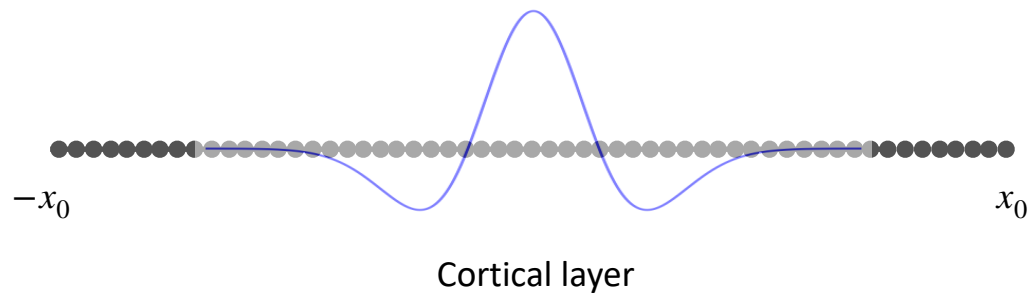
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change σ to change the range of excitation and inhibition



Amari model: synaptic interactions

Amari, Bio. Cybernetics, 1977

$$\mu \frac{\partial u(x, t)}{\partial t} = -u(x, t) + \int_{-\infty}^{\infty} s(u(y)) \omega(x - y) dy + h + P(x, t)$$

Threshold stimulus

h : constant, it determines the resting state value

- In the absence of a time-dependent external stimulus ($P(x, t) = 0$ for all $x \in [-x_0, x_0]$, $t \in [0, T]$), the type of stationary solution ($\frac{\partial u(x, t)}{\partial t} = 0$ for all $x \in [-x_0, x_0]$ and for all $t > t_0$ where $t_0 > 0$) depends on the value of h .

Amari model: equilibrium solutions

Amari, Bio. Cybernetics, 1977

See the lecture notes for types and stability of equilibrium solutions!